

JUNXIAN WU

Senior AI Engineer | Vision, Agentic & Multimodal Systems

Singapore | vemodalenx@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFILE

Senior AI Engineer with 4+ years turning computer-vision and multimodal models into deployable, measurable, and maintainable systems. Production depth in camera AI, computational photography, edge deployment, C++/Android integration, and visual-quality evaluation; extending the same systems discipline into agentic workflows, local RAG, structured outputs, provider routing, failure handling, and evaluation. Strong fit for Applied AI and Forward Deployed Engineer roles that bridge model capability, runtime constraints, and product delivery.

SYSTEM SPINE

Perception: visual and multimodal signals | **Reasoning:** rules, state, prompts, schemas | **Runtime:** native, edge, services, APIs | **Evaluation:** hard cases, traces, release gates | **Handoff:** interfaces, debug paths, operating knowledge

PROFESSIONAL EXPERIENCE

Senior AI Engineer | Black Sesame Technologies (Singapore)

Jan 2025 - Present

- Own end-to-end delivery for camera and image AI features, translating product requirements and field feedback into technical scope, prototypes, evaluation gates, and production handoff.
- Lead real-time composition and portrait-imaging systems combining detection, optical-flow tracking, segmentation and matting, monocular depth, GPU rendering, and application state behavior.
- Drive model-to-device delivery across PyTorch, quantization, TFLite/ONNX, C++/Android integration, GPU pipelines, and on-device debugging.
- Build multimodal evaluation workflows with provider routing, structured outputs, retry and fallback behavior, hard-case suites, reproducible traces, and release criteria.

AI Engineer | Black Sesame Technologies (Singapore)

Aug 2022 - Jan 2025

- Productized vision models for portrait segmentation and matting, monocular depth, object detection, and camera-oriented image processing.
- Built repeatable training-to-deployment workflows spanning data preparation, training, quantization, export validation, runtime assumptions, and model replacement checks.
- Integrated models into native C++ and Android camera pipelines with simulation, cross-platform builds, device deployment, and visual debugging.
- Developed depth-aware bokeh and portrait-rendering workflows with hard-case datasets, batch comparison, golden cases, and release-review tooling.

RESEARCH EXPERIENCE

Research Intern | National University of Singapore

Dec 2021 - Jul 2022

- Conducted computer-vision research for construction-site safety, including dataset preparation, image-classification experiments, and spatial risk analysis.
- Co-authored a peer-reviewed *Automation in Construction* paper on supervised and self-supervised feature representation learning.

Research Assistant | Shandong University

Sep 2019 - Jun 2021

- Applied computer vision and machine learning to interdisciplinary psychology research, supporting dataset development, experiments, visualization, and reporting.

SELECTED SYSTEMS

Production foundation - Vision & Edge AI: Training-to-deployment workflows, quantization and export validation, native C++/Android integration, on-device debugging, and release checks.

Differentiated domain - Computational Photography: Real-time composition, portrait segmentation and matting, monocular depth, depth-aware bokeh, GPU rendering, and visual-quality evaluation.

Growth vector - Agentic & Multimodal AI: Public work in local RAG, multimodal workflow design, structured outputs, provider routing, retries and fallbacks, traces, and evaluation.

DELIVERY & TECHNICAL LEADERSHIP

- Translate ambiguous product and image-quality issues into scoped work, hard cases, interfaces, and acceptance gates.
- Coordinate with product, QA, camera tuning, runtime, and customer-facing teams through integration and release.
- Build reproducible handoff assets around assumptions, deployment checks, debugging paths, and review criteria.

TECHNICAL STACK

- **AI and vision:** PyTorch, TensorFlow, OpenCV, segmentation, matting, monocular depth, detection, optical flow, VLM/LLM workflows, RAG
- **Runtime and deployment:** Python, C++, Android/NDK, TFLite, ONNX, ONNX Runtime, TensorRT, OpenGL/GLSL, Docker, FastAPI/Flask
- **System quality:** quantization, export validation, structured outputs, provider routing, failure handling, hard-case suites, traces, release gates

EDUCATION

Master of Technology, Intelligent Systems | NUS-ISS | 2021 - 2022 | GPA 4.33 / 5.00

Bachelor of Engineering, Software Engineering | Shandong University | 2017 - 2021 | GPA 86.86 / 100 | Outstanding Graduate

PUBLICATION

Co-author, "Automated classification of 'cluttered' construction housekeeping images through supervised and self-supervised feature representation learning," *Automation in Construction*, 2023. DOI: [10.1016/j.autcon.2023.105095](https://doi.org/10.1016/j.autcon.2023.105095)

PUBLIC EVIDENCE

[Selected Work](#) | [Agentic Systems](#) | [Technical Map](#)